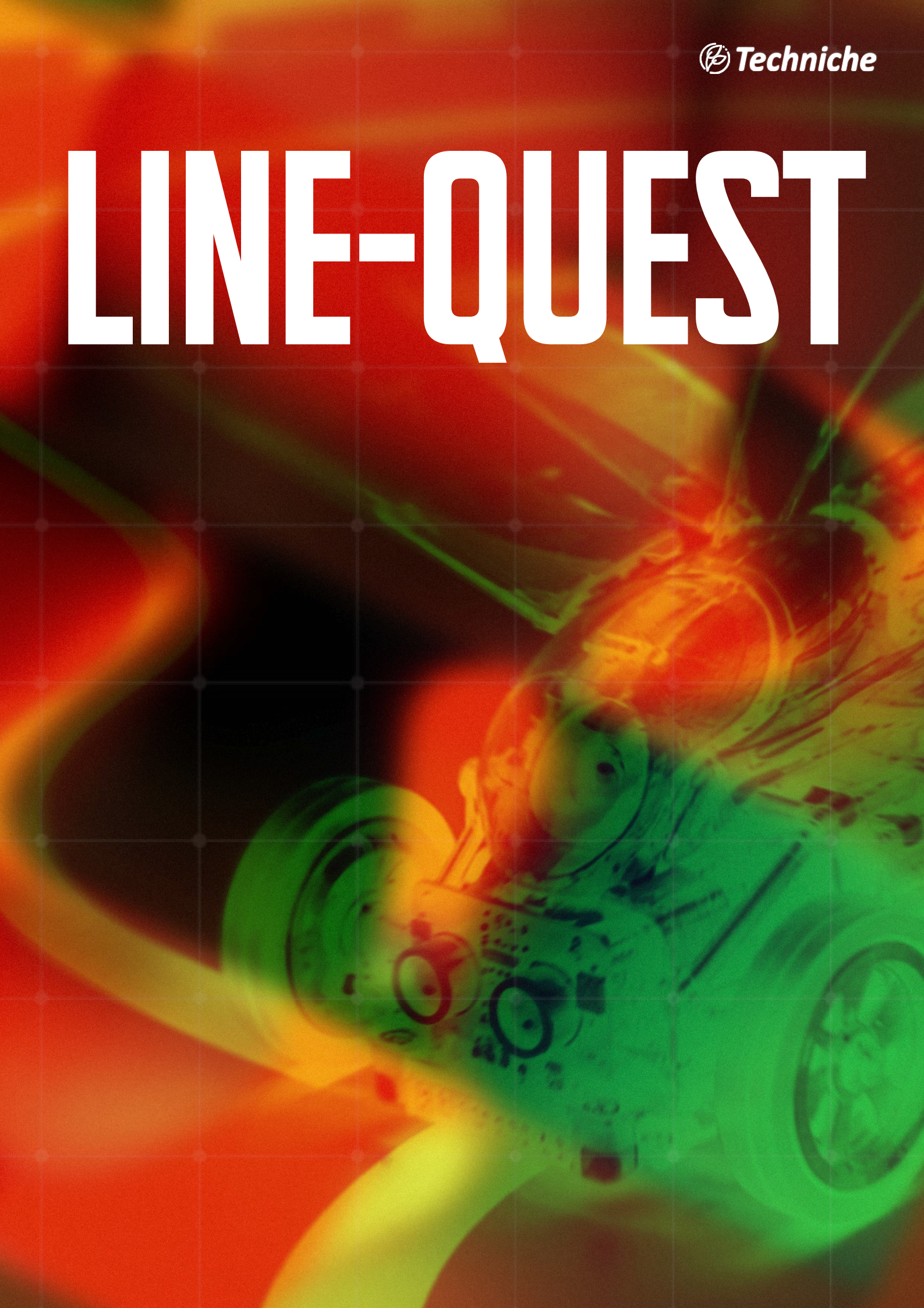
Techniche

LINE-QUEST



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AIM

Teams have to build an autonomous robot which can follow a black line and keep track of directions while going through the maze. The bot has to analyze the path in the Dry Run and use this information in the Actual Run to go through the maze from the starting point to the ending point in minimum possible time.

ARENA

- The Game field consists of 2 arenas(for 2 rounds).
- Each arena has dimensions 15ft * 7.5ft
- The arenas are composed of random paths made of black lines.
- Every curve part of track (of arena-2) is either a quarter circle or semicircle (maybe a circle too).
- The width of the track is 30mm (±5% tolerance).
- The arena will have checkpoints at random points on the track. In case the bot deviates off-track, it can start from the nearest checkpoint it previously crossed.

GAMEPLAY

The Gameplay consists of 2 parts :

1. Dry Run

In this run, the bot must start from the ‘Start Point’ and find its way to reach the ‘End Point’ of the arena. The bot has to follow an algorithm to find its path to reach ‘End’ and the bot can store the turns in its memory to explore the shortest path during the second part of the journey. There are no restrictions to cover all the checkpoints.

2. Actual Run

- In this run, the bot has to restart from the ‘Start Point’ again and finds its way to the ‘End’ through the best possible path by following the path that was stored in the first run.The ‘End Zone’ that indicates the end of the path for the bot. The timer will be set to zero as the “Actual Run” begins.
- A total of 5 minutes will be provided to complete the Dry Run. A total of 3 minutes will be provided to complete the Actual Run.
- If the bot takes more than 5 minutes to complete the dry run, then the extra time taken will be deducted from the timing of the actual run which is 3 minutes.

BOT SPECIFICATION

- The bot should fit in a cube of 250mm (±5% tolerance).
- The robot must be started by a single switch. However a team can have an onboard switch for restart.
- Bot may have a LED which will glow once it reaches the end zone of the arena or it can just beep. (not compulsory but it will give you some extra points)
- Bot must have an ‘on board’ power supply. When using the electric power supply, the potential difference between any 2 points must not exceed 24 V at any point of time during the game.
- The autonomous bot should not separate or split into two or more units.
- The autonomous bot cannot be constructed using readymade ‘Lego kits’ or any readymade mechanism. But they can make use of readymade gear assemblies. Violating this clause will lead to disqualification of the team.

GAME RULES

- Only one autonomous bot per team is allowed.
- Teams will be given 1 minute for calibration. If any team is found to alter its code after depositing its bots, then it will be immediately disqualified from the competition. They are, however, allowed to make any other hardware changes.
- When the autonomous bot starts, no team member is allowed to touch the bot.(If there is any manual intervention then there will be a penalty of 5 points)
- At the start of the task, the bot will be placed at the starting point. Only 1 team member is allowed to be near the game field while starting the bot.
- The run will start only when organizers give the signal.
- The starting procedure of the bot should be simple and should not involve giving the bot any manual force or impulse in any direction.
- A total of 10 minutes will be given. 5 minutes for Dry run, 3 minutes for Actual run and 2 minute break between both runs.
- Only 1 member of the team is allowed to handle the bot.
- Participants are not allowed to keep anything inside the arena other than the bot.
- The time measured by the organizers will be final and will be used for scoring the teams.
- Time measured by any contestant by any other means is not acceptable for scoring.
- In case of any disputes/discrepancies, the organizer's decision will be final and binding.
- The organizers reserve the right to change any or all of the above rules as they deem fit..
- Change in rules, if any will be notified to the registered teams in prior.
- The contestant must not alter the bot in any manner that reduces its weight (e.g. removal of a bulky sensor array or switching to lighter batteries to get better speed).The organizers reserve the right to arbitrate in such circumstances.

<u>Restart</u>

- If the bot takes a restart in the first part (Dry Run) of the competition, it has to start from the checkpoint.
- Once Dry Run is completed by the bot, the team won't be given any other chance for Dry Run.
- If the bot takes a restart in the second part (Actual Run) of the competition, it has to start from the start zone of the arena.
- The timer will not be set back to zero and will not be paused in any case.
- During a restart, a contestant must not feed information about the arena to the bot. However, contestants are allowed to adjust sensors (gain, position etc.) and make hardware changes.

JUDGING CRITERIA

- 25 points as it crosses any of the checkpoints but it will be counted only once for each checkpoint in Dry run.
- 40 points if the bot successfully completes the Dry Run.
- 20 points if the bot successfully completes the Actual Run.
- 10 points if the bot glows the LED upon reaching the End Zone.

SCORING

- A = 25 points × (Number of checkpoints covered during the Dry Run)
- B = 40 points if the bot successfully completes the Dry Run
- C = 300 - Total time taken in seconds in completing the Dry Run.
- D= 20 points if the bot successfully completes the Actual Run.
- T = 200 - Total time taken to complete the Actual Run.
- L = 10 points if the LED glows upon reaching the End Zone. This LED should not glow in between the Dry/ Actual Run.
- P = Penalties
- Total = (A + B + C + L + T) - P

- Team Total = Round 1 Total + Round 2 Total

TEAM CRITERIA

- A team may consist of a Maximum of 5 members.
- Students from different educational institutions are allowed to form a team.
- All students must have a valid identity card of their respective educational institution.

REGISTRATION

1. The team leader of every team can register on Unstop for free or directly through the Techniche website to take part in Line Quest.
2. Upon registering in unstop, the team leader will be emailed the further procedure of registration on the official Techniche Portal(techniche.org).
3. The team leader must then register their team on the Techniche portal and pay the registration fee of Rs. 200/-.
4. All team members of the team are required to register on Techniche to obtain their individual Techniche IDs.
5. If you are a team leader, proceed to fill out the Line Quest Registration form (enter only the team leader's name).
6. If you are not a Team Leader, just register on Techniche to get your Techniche ID and share it with your Team Leader to become a part of the Team.
7. Team Leaders are requested to visit the ‘Add Member’ and enter the unique Techniche ID of your colleagues you want to add to your team(Your colleague must be registered on Techniche and have a unique Techniche ID).
8. An invitation link will be sent to your colleague's email ID. A colleague will become a confirmed member of your team upon accepting the invitation.

CONTACT US

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